MLJ Contribution Information Sheet

FALI: Mechanistic Decomposition of Conditional Attention-Head Importance

1. What is the main claim, and why is it important to machine learning?

A transformer's component importance can change after another circuit component is perturbed, but identifying that conditional change does not explain its mechanism. We introduce FALI (Factorized Attention-Level Intervention), a hypothesis-driven framework for a selected attention head that decomposes the cross-state change in its output-grounded causal usefulness into target scores (T), rival scores (R), other valid attention scores (O), value content (V), and surrounding-network context (C). Standard Owen attribution allocates interactions under the fixed coalition partition $\{T, R, O\}$, $\{V\}$, $\{C\}$ and reconciles the fine score decomposition with the coarse score/value/context game. The Owen mathematics is established; the contribution is the intervention-defined estimand and architecture-aligned causal game.

This distinction matters because a shared functional label can conceal different internal mechanisms. In our experiments, GPT-2 IOI and Pythia induction both exhibit positive conditional recruitment, yet their target/rival score contributions have opposite signs. Thus, detecting that a component became important is not sufficient to infer how its importance changed.

2. What precise evidence supports the claim?

1. **Controlled validation.** Five deterministic systems isolate T, R, O, V, and C one at a time; FALI assigns the endpoint change to the manipulated channel with maximum discrepancy 1.11×10^{-16}. A pure $T \times C$ interaction is split symmetrically by standard Owen attribution. A randomly initialized two-layer transformer independently verifies structural masks, score-to-pattern and pattern/value-to-head-output reconstruction, direct endpoint identities, candidate-isolation, live-Q/V leakage checks, and coarse–fine reconciliation, with zero recorded model-level discrepancies at the preregistered precision.
2. **GPT-2 IOI.** On 15 canonical IOI templates crossed with BABA/ABBA patterns, four prompt seeds, and 3600 prompts, L10H2 is positively recruited after ablating the canonical primary Name Movers: conditional gain 0.298 (95% hierarchical CI $[0.263, 0.335]$). Its FALI decomposition gives $\phi_T = 0.783$ ($[0.722, 0.838]$) and $\phi_R = -0.341$ ($[-0.372, -0.311]$). L10H10 provides a functional contrast: target attention increases strongly, but conditional gain is negative (-0.263, CI $[-0.302, -0.225]$). The four preregistered GPT-2 fine-channel hypotheses pass across the broad canonical design and the template-overlap-excluded sensitivity.
3. **Intervention checks.** The target-positive/rival-negative GPT-2 channels retain their directions under the tested matched-mean and matched-resample primary-damage schemes. We do not report a blanket robustness claim: the preregistered EF-5 experiment-level gate failed because its zero-damage reference criterion was not satisfied.
4. **Preregistered cross-setting test.** In Pythia-160M induction, a discovery procedure that was forbidden from inspecting FALI channels selected L5H0→L8H2; confirmation used held-out seeds 1, 8, and 22. L8H2 is positively recruited: conditional effect 0.621 ($[0.456, 0.860]$), conditional gain 0.486 ($[0.326, 0.722]$). The preregistered GPT-2 mechanism does not transfer: $\phi_T = -0.144$ ($[-0.247, -0.084]$) and $\phi_R = +0.118$ ($[0.058, 0.219]$). The EF-6 gate therefore fails specifically because both fine-score directions reverse while recruitment remains confirmed.

3. Which papers are closest, and how is this work related?

The closest lines of work are: (i) self-repair and backup-head studies, including the Hydra Effect, Copy Suppression, and broader self-repair analyses; (ii) Conditional Co-Ablation (CoAx), which measures which components become more important after a primary set is removed; (iii) interaction-aware attribution, especially Causal Head Gating and *The Curse of Multiple Mediators*, which establish dependence of component effects on the surrounding model; (iv) activation/path patching; (v) QK/OV and recent causal routing-versus-value decompositions; (vi) prompt-specific circuit analyses, which show systematic mechanism variation across IOI prompt families and model families; (vii) Epistatic Circuits, which exhaustively decomposes higher-order interactions among circuit components; and (viii) divergent/off-manifold intervention analyses, which caution that counterfactual internal states need not lie on the model's natural manifold.

FALI is positioned downstream of these methods. It does not claim first discovery of self-repair, conditional importance, context-dependent component roles, routing/value separation, prompt-specific heterogeneity, epistatic interaction structure, or Shapley/Owen attribution. Given a selected attention head whose causal usefulness changes across two model states, FALI asks which architecture-level changes account for that change under an explicit counterfactual game. To our knowledge, the closest prior work does not provide this exact output-grounded cross-state decomposition into score/value/context with target/rival/other-score resolution. The manuscript states this novelty claim narrowly and immediately cites the closest conditional-attribution, interaction, routing/value, prompt-specific, epistatic, and off-manifold work.

4. Have parts appeared previously? What is the significant contribution beyond them?

An earlier public preprint, *When Attention Repairs Itself: A Mechanistic Study of Backup Heads in GPT-2* (May 2026; DOI `10.5281/zenodo.20350000`), used the earlier name "Frozen Attention Logit Intervention" and centered an attention-based interpretation of GPT-2 backup behavior. Corrected functional analysis subsequently falsified the assumption that increased target attention implied increased causal usefulness.

The present work is a scientific reconstruction rather than a cosmetic extension. It changes the estimand from attention change to output-grounded conditional causal usefulness; replaces the earlier two-score analysis with the five-factor Factorized Attention-Level Intervention game; corrects the attribution rule to standard Owen values; adds synthetic and random-transformer ground-truth validation; uses the corrected canonical three-Name-Mover perturbation; confirms fine-score effects across broad IOI templates and multiple seeds; tests alternative damage schemes; and performs a preregistered Pythia induction transfer. That transfer rejects the proposed GPT-2 fine-score generalization by producing opposite target/rival signs. The current manuscript therefore supersedes the earlier mechanistic interpretation and explicitly reports the negative results that motivated the reconstruction.